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Open Geospatial Consortium

OGC EMISSION EVENT MODELING LANGUAGE (EMISSIONML) PART I: DATA MODEL

DRAFT STANDARD

DRAFT

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I KEYWORDS

The following are keywords to be used by search engines and document catalogues.

ogcdoc, OGC document, emissions, climate change

II PREFACE

i. Abstract

This OGC Standard defines EmissionML, a conceptual data model for representing atmospheric emission events. EmissionML provides a domain-specific ontology for describing the release of substances into the atmosphere, including the source from which an emission originates, the mechanism by which it occurs, the quantity of substance released, and the temporal bounds of the event.

EmissionML distinguishes itself through three key design principles. First, it incorporates an evidence chain linking emission determinations to supporting Observations, enabling traceability from reported values to underlying measurements. Second, it provides explicit uncertainty provenance by attaching data quality metadata to both emission quantities and temporal bounds, ensuring that the confidence and precision of each determination are recorded alongside the values themselves. Third, it offers clear spatial and temporal context — spatial through Feature (as the ultimate feature of interest) and Source (as the emission origin), temporal through start time, end time, and the phenomenon time of supporting Observations. By adopting the OMS 3.0 Observation model, EmissionML provides a common framework for reconciling heterogeneous sensor observations — from remote sensing satellites to ground-based continuous monitoring stations — within a single, unified evidence chain.

The model builds on foundational OGC and ISO standards. Source inherits from Feature (ISO 19109), DeterminationProcedure inherits from Procedure (ISO/OGC 19156:2023), and Observations conform to the OMS 3.0 Observation model (ISO/OGC 19156:2023). The standard defines twelve requirements specifying the structure and constraints of the core data model.

By making emission semantics, evidence chains, and uncertainty provenance explicit and machine-verifiable, EmissionML enables credibility of reported emission values, traceability from determinations to their supporting observations and procedures, and reproducibility of emission assessments across organizations and jurisdictions.

The intended audience includes environmental regulators, greenhouse gas reporting frameworks, emissions monitoring system developers, and geospatial data infrastructure implementers who require a standardized, interoperable representation of emission event data.

ii. Keywords

The following are keywords to be used by search engines and document catalogues.

ogcdoc, OGC document, <tags separated by commas>

iii. Preface

NOTE: Insert Preface Text here. Give OGC specific commentary: describe the technical content, reason for document, history of the document and precursors, and plans for future work. > Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. The Open Geospatial Consortium shall not be held responsible for identifying any or all such patent rights.

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iv. Security Considerations

No security considerations have been made for this Standard.

v. Submitting organizations

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

Organization name(s)

vi. Submitters

All questions regarding this submission should be directed to the editor or the submitters:

Table 1

Name	Affiliation
Steve Liang	University of Calgary / SensorUp
Ryan Ahola	Natural Resources Canada

III SECURITY CONSIDERATIONS

No security considerations have been made for this document.

IV SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

- SensorUp Inc.
- Natural Resources Canada
- University of Calgary

1

SCOPE

1 SCOPE

NOTE: Insert Scope text here. Give the subject of the document and the aspects of that scope covered by the document.

2

NORMATIVE REFERENCES

2 NORMATIVE REFERENCES

There are no normative references in this document.

The following normative documents contain provisions that, through reference in this text, constitute provisions of this document. For dated references, subsequent amendments to, or revisions of, any of these publications do not apply. For undated references, the latest edition of the normative document referred to applies.

NOTE: Insert References here. If there are no references, state "There are no normative references".

References are to follow the Springer LNCS style, with the exception that optional information may be appended to references: DOIs are added after the date and web resource references may include an access date at the end of the reference in parentheses. See examples from Springer and OGC below.

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ISO / TC 211: ISO 19115-1:2014 Geographic information — Metadata — Part 1: Fundamentals (2014)

ISO / TC 211: ISO 19157:2013 Geographic information — Data quality (2013)

ISO / TC 211: ISO 19139:2007 Geographic information — Metadata — XML schema implementation (2007)

ISO / TC 211: ISO 19115-3: Geographic information — Metadata — Part 3: XML schemas (2016)

OGC: OGC 15-097 OGC Geospatial User Feedback Standard. Conceptual Model (2016)

OGC: OGC 12-019, OGC City Geography Markup Language (CityGML) Encoding Standard (2012)

OGC: OGC 14-005r3, OGC IndoorGML (2014)

3

TERMS AND DEFINITIONS

3 TERMS AND DEFINITIONS

No terms and definitions are listed in this document.

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word "shall" (not "must") is the verb form used to indicate a requirement to be strictly followed to conform to this Standard and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications (OGC 08-131r3), also known as the 'ModSpec'. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

For the purposes of this document, the following additional terms and definitions apply.

Determination Procedure	A Determination Procedure describes the steps performed to determine the quantity or temporal bounds of an Emission Event.
Emission Event	An Emission Event represents the release of a substance into the atmosphere over a non-zero duration. It is characterized by its start time and end time, the Source Feature from which the substance originated, and the identity and quantity of the emitted substance.
Emission Quantity	An Emission Quantity is the physical quantity representing the total mass of substance released to the atmosphere during a defined Emission Event or over a specified time interval.
Emission Intent	Emission Intent is a code list describing whether an emission is intentional, unintentional, or unknown.
Feature	Abstraction of real world phenomena (Source: ISO 19101-1:2014)
Measurement	Set of operations having the object of determining the value of a quantity (Source: ISO/OGC 19156:2023)
Mechanism	A Mechanism is a classification of the specific physical, chemical, or operational process that constitutes the direct cause or pathway for an emission. It is defined independently of the Source—including its identity, structure, or purpose—allowing Mechanisms to be consistently reused across different types of Source.
Observation	Act carried out by an observer to determine the value of an observable property of an object (feature-of-interest) by using a procedure, with the value provided as the result (Source: ISO/OGC 19156:2023)
Procedure	Description of steps performed (Source: ISO/OGC 19156:2023)
Source	A Source represents the feature from which an emission originates. A Source is an ISO 191xx feature whose geometry provides the geospatial reference for one or more Emission Events. A Source is

	characterised by its own identity, geometry, intended function, and lifecycle, and is explicitly agnostic to the Mechanism by which any associated Emission Event occurs.
Source Type	A Source Type classifies a Source by its category.
Temporal Bound	A Temporal Bound is a value object representing a temporal determination (e.g., start time or end time) that MAY have associated data quality metadata.
Unit of Measure	Reference quantity chosen from a unit equivalence group (Source: ISO/OGC 19156:2023)

4

CLAUSES NOT CONTAINING NORMATIVE MATERIAL

4 CLAUSES NOT CONTAINING NORMATIVE MATERIAL

Paragraph

4.1. Clauses not containing normative material sub-clause 1

Paragraph

4.2. Clauses not containing normative material sub-clause 2

5

EMISSION EVENT DATA MODEL

5 EMISSION EVENT DATA MODEL

The EmissionML conceptual model is described as a class diagram in Figure 1. The schema is fully described in x.x.x.

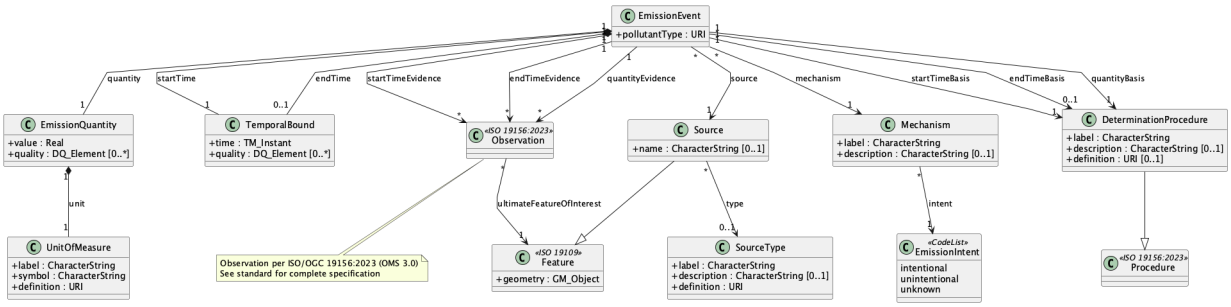
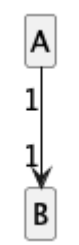


Figure 1 — EmissionML Abstract Conceptual Model

Table 2 — UML relationship notation used in this standard

SYMBOL	NAME	MEANING
	Composition	The target is owned by and cannot exist without the source (e.g., EmissionEvent owns its EmissionQuantity and TemporalBound instances)
	Association	A directional reference between independent classes (e.g., EmissionEvent refers to a Source)

Use the following table for Requirements Classes.

Table 3

Emission Event Conceptual Data Model Core	
http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core	
Target type	Data Model

Dependency	http://www.opengis.net/doc/as/om/3.0/req/obs-basic
Requirement 1	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/emission-event This requirement defines the Emission Event.
Requirement 2	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/source This requirement defines the Source.
Requirement 3	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/source-type This requirement defines the Source Type.
Requirement 4	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/mechanism This requirement defines the Mechanism.
Requirement 5	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/emission-quantity This requirement defines the Emission Quantity.
Requirement 6	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/unit-of-measure This requirement defines the Unit of Measure.
Requirement 7	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/temporal-bound This requirement defines the Temporal Bound.
Requirement 8	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/observation This requirement defines the Observation.
Requirement 9	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/emission-intent This requirement defines the Emission Intent.
Requirement 10	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/determination-procedure This requirement defines the Determination Procedure.
Requirement 11	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/feature This requirement defines the Feature.
Requirement 12	http://www.opengis.net/spec/emissionml/1.0/req/datamodel/core/procedure This requirement defines the Procedure.

5.1. Emission Event Class

An Emission Event represents the release of a substance into the atmosphere over a non-zero duration. It is characterized by its start time and end time, the Source Feature from which the substance originated, and the identity and quantity of the emitted substance.

NOTE 1: Conceptually, an Emission Event corresponds to an occurrent in upper-ontology terms, as it unfolds over time and has temporal parts.

NOTE 2: For reporting or accounting purposes, a long-running emission event is often segmented into multiple time-bounded emission events (e.g., annual records). This segmentation reflects reporting conventions and does not imply physical cessation or re-initiation of the emission.

Table 4

Requirement 1: Emission Event Class	/req/datamodel/core/emission-event Each Emission Event class shall have the mandatory attributes listed in Table 5. Each Emission Event class shall have the direct relation between an Emission Event class and other classes listed in Table 6.

Table 5 — Attributes of an Emission Event class

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
pollutant Type	The URI linking to an external definition of the pollutantType of the Emission Event.	URI	1 (mandatory)

Table 6 — Direct relation between an Emission Event class and other class types

ROLE	DESCRIPTION	TARGET CLASS TYPE	MULTIPLICITY AND USE
hasMechanism	The mechanism by which the emission occurred.	Mechanism	1 (mandatory)
hasSource	The source from which the emission originated.	Source	1 (mandatory)
startTime	The start time of the emission event.	TemporalBound	1 (mandatory)
endTime	The end time of the emission event.	TemporalBound	0..1 (optional)
hasQuantity	The quantity of the substance emitted.	EmissionQuantity	1 (mandatory)
hasStartTimeBasis	The procedure used to determine the start time.	Determination Procedure	1 (mandatory)
hasEndTimeBasis	The procedure used to determine the end time.	Determination Procedure	0..1 (optional)
hasQuantityBasis	The procedure used to determine the emission quantity.	Determination Procedure	1 (mandatory)
hasStartTime Evidence	Observations providing evidence for the start time determination.	Observation	0..* (optional)

ROLE	DESCRIPTION	TARGET CLASS TYPE	MULTIPLICITY AND USE
hasEndTimeEvidence	Observations providing evidence for the end time determination.	Observation	0..* (optional)
hasQuantityEvidence	Observations providing evidence for the quantity determination.	Observation	0..* (optional)

5.2. Feature Class

A Feature is an abstraction of real world phenomena (Source: ISO 19101-1:2014). In EmissionML, Feature serves as the base class for entities that can be the ultimate feature of interest of an Observation. A Feature's `geometry` attribute provides the geospatial reference for the phenomenon it represents.

Table 7

Requirement 11: Feature Class	/req/datamodel/core/feature + Each Feature class shall have the mandatory attributes listed in Table 8.

Table 8 — Attributes of a Feature class

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
geometry	The spatial geometry of the feature.	GM_Object	1 (mandatory)

Table 9 — Direct relation between a Feature class and other class types

ROLE	DESCRIPTION	TARGET CLASS TYPE	MULTIPLICITY AND USE
-	No direct relations are defined for Feature in the core model (it is referenced by Source via inheritance and by Observation via ultimateFeatureOfInterest).	-	-

5.3. Source Class

A Source represents the feature from which an emission originates. A Source is an ISO 191xx feature whose geometry provides the geospatial reference for one or more Emission Events. A

Source is characterised by its own identity, geometry, intended function, and lifecycle, and is explicitly agnostic to the Mechanism by which any associated Emission Event occurs.

NOTE: A Source is a continuant that may exist independently of any emission, whereas an Emission Event is an occurrent. The identity and lifecycle of a Source are independent of the existence, absence, or frequency of Emission Events.

Table 10

Requirement 2: Source Class	/req/datamodel/core/source + Each Source class shall inherit from Feature. Each Source class shall have the optional attributes listed in Table 11. Each Source class shall have the direct relation between a Source class and other classes listed in Table 12.

Table 11 — Attributes of a Source class (own attributes; geometry is inherited from Feature)

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
name	A human-readable name for the source.	CharacterString	0..1 (optional)

Table 12 — Direct relation between a Source class and other class types

ROLE	DESCRIPTION	TARGET CLASS TYPE	MULTIPLICITY AND USE
hasType	The type classification of the source.	SourceType	0..1 (optional)

5.4. Source Type Class

A Source Type classifies a Source by its category.

Table 13

Requirement 3: Source Type Class	/req/datamodel/core/source-type + Each Source Type class shall have the mandatory attributes listed in Table 14.

Table 14 — Attributes of a Source Type class

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
label	A human-readable label for the source type.	CharacterString	1 (mandatory)
description	A detailed description of the source type.	CharacterString	0..1 (optional)
definition	The URI linking to an external definition of the source type.	URI	1 (mandatory)

Table 15 — Direct relation between a Source Type class and other class types

ROLE	DESCRIPTION	TARGET CLASS TYPE	MULTIPLICITY AND USE
-	No direct relations are defined for SourceType in the core model (it is referenced by Source).	-	-

5.5. Mechanism Class

A Mechanism is a classification of the specific physical, chemical, or operational process that constitutes the direct cause or pathway for an emission. It is defined independently of the Source—including its identity, structure, or purpose—allowing Mechanisms to be consistently reused across different types of Source.

Table 16

Requirement 4: Mechanism Class	/req/datamodel/core/mechanism + Each Mechanism class shall have the mandatory attributes listed in Table 17. Each Mechanism class shall have the direct relation listed in Table 18.

Table 17 — Attributes of a Mechanism class

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
label	A human-readable label for the mechanism.	CharacterString	1 (mandatory)
description	A detailed description of the mechanism.	CharacterString	0..1 (optional)

Table 18 — Direct relation between a Mechanism class and other class types

ROLE	DESCRIPTION	TARGET CLASS TYPE	MULTIPLICITY AND USE
has Intent	The emission intent associated with the mechanism.	EmissionIntent	1 (mandatory)

5.6. Emission Quantity Class

An Emission Quantity is the physical quantity representing the total mass of substance released to the atmosphere during a defined Emission Event or over a specified time interval.

Table 19

Requirement 5: Emission Quantity Class	/req/datamodel/core/emission-quantity + Each Emission Quantity class shall have the mandatory attributes listed in Table 20. Each Emission Quantity class shall have the direct relation between an Emission Quantity class and other classes listed in Table 21.

Table 20 — Attributes of an Emission Quantity class

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
value	The numeric value of the emission quantity.	Real	1 (mandatory)
quality	A data quality element describing the quality of the emission quantity value.	DQ_Element	0..* (optional)

Table 21 — Direct relation between an Emission Quantity class and other class types

ROLE	DESCRIPTION	TARGET CLASS TYPE	MULTIPLICITY AND USE
has Unit	The unit of measure associated with the emission quantity.	UnitOfMeasure	1 (mandatory)

5.7. Unit of Measure Class

A Unit of Measure is a reference quantity chosen from a unit equivalence group (Source: ISO/OGC 19156:2023)

Table 22

Requirement 6: Unit of Measure Class	/req/datamodel/core/unit-of-measure + Each Unit of Measure class shall have the mandatory attributes listed in Table 23.

Table 23 — Attributes of a Unit of Measure class

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
label	A human-readable label for the unit of measure.	CharacterString	1 (mandatory)
symbol	A short symbol representing the unit of measure.	CharacterString	1 (mandatory)
definition	The URI linking to an external definition of the unit of measure.	URI	1 (mandatory)

Table 24 — Direct relation between a Unit of Measure class and other class types

ROLE	DESCRIPTION	TARGET CLASS TYPE	MULTIPLICITY AND USE
-	No direct relations are defined for UnitOfMeasure in the core model.	-	-

5.8. Temporal Bound Class

A Temporal Bound is a value object representing a temporal determination (e.g., start time or end time) that MAY have associated data quality metadata. The determination procedure and supporting evidence for a Temporal Bound are tracked at the Emission Event level.

Table 25

Requirement 7: Temporal Bound Class	/req/datamodel/core/temporal-bound + Each Temporal Bound class shall have the mandatory attributes listed in Table 26.

Table 26 — Attributes of a Temporal Bound class

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
time	The time instant value of this Temporal Bound.	TM_Instant	1 (mandatory)
quality	A data quality element describing the quality of the temporal bound.	DQ_Element	0..* (optional)

Table 27 — Direct relation between a Temporal Bound class and other class types

ROLE	DESCRIPTION	TARGET CLASS TYPE	MULTIPLICITY AND USE
-	No direct relations are defined for TemporalBound in the core model (determination procedure and evidence are tracked at the EmissionEvent level).	-	-

5.9. Observation Class

An Observation is an act carried out by an observer to determine the value of an observable property of an object (feature-of-interest) by using a procedure, with the value provided as the result (Source: ISO/OGC 19156:2023). In EmissionML, an Observation is evidence that may support an Emission Quantity or Temporal Bound determination.

NOTE: The Ultimate Feature of Interest of an Observation identifies a Feature — which may be a Source or another feature at a different spatial scale — associated with the Emission Event. It is common for a single Emission Event to be observed by multiple Observations whose Ultimate Features of Interest are defined at different spatial scales. For example, a satellite remote sensing image Observation typically has an area-based Ultimate Feature of Interest (e.g., a facility area or region), whereas the same Emission Event may also be observed by a person carrying an Optical Gas Imaging camera, for which the Ultimate Feature of Interest is a specific Source such as a component of an item of equipment from which pollutants are released.

Table 28

Requirement 8: Observation Class	/req/datamodel/core/observation + Each Observation class shall be an ISO/OGC 19156:2023 (OGC OMS 3.0) Observation.

5.10. Emission Intent Enumeration

Emission Intent is a code list describing whether an emission is intentional, unintentional, or unknown.

Table 29

Requirement 9: Emission Intent Enumeration	/req/datamodel/core/emission-intent + The Emission Intent enumeration shall support the enumeration values listed in Table 30.

Table 30 — Enumeration values of Emission Intent

NAME	DEFINITION
intentional	The emission occurred intentionally.
unintentional	The emission occurred unintentionally.
unknown	The intent of the emission is unknown.

5.11. Procedure Class

A Procedure is a description of steps performed (Source: ISO/OGC 19156:2023). In EmissionML, Procedure serves as the base class for process descriptions such as DeterminationProcedure.

Table 31

Requirement 12: Procedure Class	/req/datamodel/core/procedure + Each Procedure class shall be an ISO/OGC 19156:2023 (OGC OMS 3.0) Procedure.

Table 32 — Attributes of a Procedure class

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
-	No own attributes are defined for Procedure in the core model (see the standard for complete specification).	-	-

Table 33 — Direct relation between a Procedure class and other class types

ROLE	DESCRIPTION	TARGET CLASS TYPE	MULTIPLICITY AND USE
-	No direct relations are defined for Procedure in the core model (it is referenced by DeterminationProcedure via inheritance).	-	-

5.12. Determination Procedure Class

A Determination Procedure describes the steps performed to determine the quantity or temporal bounds of an Emission Event. DeterminationProcedure inherits from Procedure, aligning it with the OMS 3.0 process model.

Table 34

Requirement 10: Determination Procedure Class	/req/datamodel/core/determination-procedure + Each Determination Procedure class shall inherit from Procedure. Each Determination Procedure class shall have the mandatory attributes listed in Table 35.

Table 35 — Attributes of a Determination Procedure class (own attributes)

NAMES	DEFINITION	DATA TYPES AND VALUES	MULTIPLICITY AND USE
label	A human-readable label for the determination procedure.	CharacterString	1 (mandatory)
description	A detailed description of the determination procedure.	CharacterString	0..1 (optional)
definition	The URI linking to an external definition of the determination procedure.	URI	0..1 (optional)

A

ANNEX A (INFORMATIVE) RELATIONSHIP WITH OGC FEATURES AND OBSERVATIONS

A ANNEX A (INFORMATIVE) RELATIONSHIP WITH OGC FEATURES AND OBSERVATIONS

EmissionML is a domain-specific conceptual model that builds on two foundational OGC concepts: **Features** (ISO 19109 General Feature Model) and **Observations** (ISO/OGC 19156:2023, OMS 3.0). This annex explains how EmissionML relates to each and why explicit domain modeling is preferred over generic usage.

Figure A.1 illustrates the conceptual layering of EmissionML on top of these foundational standards.

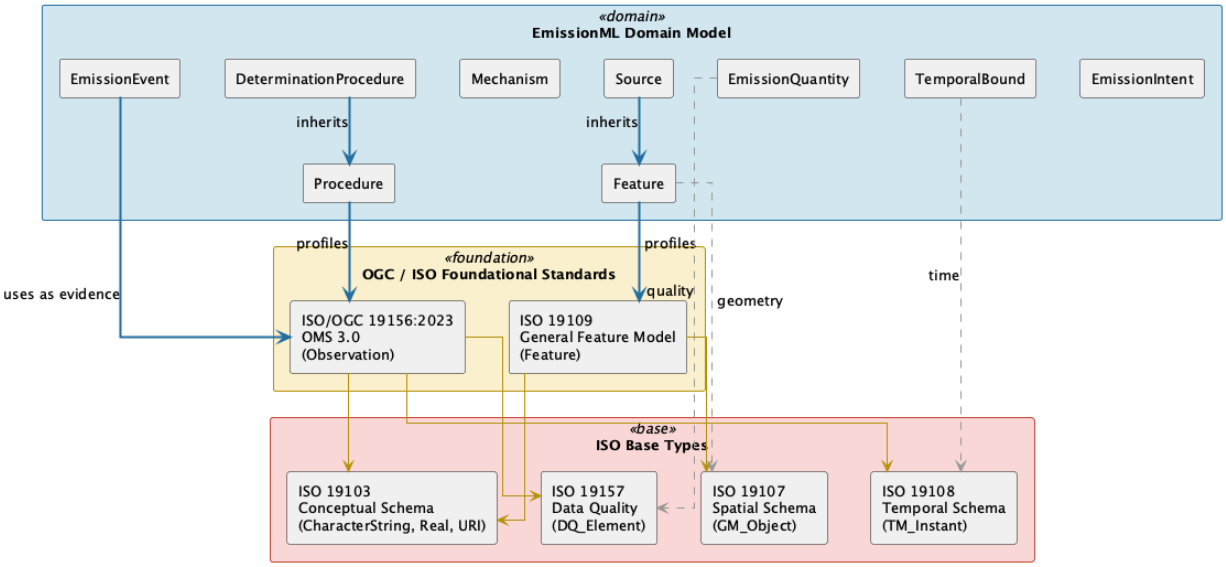


Figure A.1 — Conceptual layering of EmissionML on foundational ISO/OGC standards

A.1. EmissionML and OGC Features

The OGC General Feature Model (ISO 19109) defines a *Feature* as an abstraction of a real-world phenomenon — any entity with an identity, properties, and optionally a geometry. EmissionML makes this alignment explicit by introducing a concrete **Feature** class (stereotyped `[ISO_19109]`) into the conceptual model.

The Feature class carries a mandatory `geometry` attribute of type `GM_Object` and serves as the base class for domain-specific feature types. **Source** inherits from Feature, gaining its geometry by inheritance. This structural inheritance — rather than a notional stereotype — makes the ISO 19109 alignment unambiguous and machine-verifiable.

Because Feature is concrete, an Observation's `ultimateFeatureOfInterest` can reference any Feature — not only a Source. This enables observations whose feature of interest is defined at a different spatial scale (e.g., a facility area, a region, or a monitoring zone) to participate in the evidence chain without being forced into the Source class.

EmissionEvent is also a real-world phenomenon — an occurrent that unfolds over time. In principle, an Emission Event *could* be modeled as a generic Feature. However, EmissionML gives it explicit structure: temporal bounds (start time, end time), a quantity of emitted substance, a mechanism describing the physical or operational process, an intent classification, and an evidence chain linking the event to supporting Observations.

Without this explicit modeling, a Source would be indistinguishable from any other Feature, and an emission event would be "just another Feature" with ad-hoc properties. The semantic relationships that give emission data its meaning — `hasSource`, `hasMechanism`, `hasQuantity`, `hasStartTimeEvidence` — would be lost or require convention-based interpretation.

A.2. EmissionML and OMS 3.0 Observations

OMS 3.0 (ISO/OGC 19156:2023) defines an Observation as an act of determining the value of an observable property. An Observation has a well-defined structure: a feature of interest, an observed property, a procedure, and a result.

EmissionML incorporates OMS 3.0 Observation **by reference** (Requirement 8) — it does not redefine or extend the Observation class. In EmissionML, Observations serve as **evidence** supporting emission quantity and temporal bound determinations, linked through the `hasQuantityEvidence`, `hasStartTimeEvidence`, and `hasEndTimeEvidence` associations.

One might ask: *Why not model the entire emission event as an Observation result?* Because an Emission Event is richer than what a single Observation captures. An Emission Event has a source, a mechanism, an intent, multiple temporal bounds, and can be supported by *multiple* Observations at different spatial and temporal scales. For example:

- A satellite remote sensing Observation may detect a methane plume over a broad area.
- An Optical Gas Imaging (OGI) camera Observation may pinpoint the leak to a specific component.
- A continuous emissions monitoring system (CEMS) Observation may quantify the emission rate over time.

All three Observations can serve as evidence for a single Emission Event, each contributing to the determination of quantity, start time, or end time. The Observation's `ultimateFeatureOfInterest` references a Feature — which may be a Source or any other

Feature at a different spatial scale. The satellite Observation's feature of interest may be a Feature representing an entire facility, while the OGI Observation's feature of interest is a Source representing a single valve.

A.3. EmissionML and OMS 3.0 Procedures

OMS 3.0 defines a **Procedure** as "a description of steps performed" and introduces **ObservingProcedure** as a specialization that describes steps performed to determine the value of an observable property. EmissionML follows this same specialization pattern.

The EmissionML conceptual model introduces a **Procedure** class (stereotyped `ISO 19156/2023`) as the base class for process descriptions. **DeterminationProcedure** inherits from Procedure and describes the steps performed to determine a value — such as an emission quantity or temporal bound. This structural inheritance mirrors how ObservingProcedure specializes Procedure in OMS 3.0.

By making Procedure an explicit base class rather than a notional reference, EmissionML ensures that its process descriptions are structurally aligned with OMS 3.0 — not merely by naming convention, but by class hierarchy. This parallels the Feature/Source pattern: just as Source inherits from Feature to make ISO 19109 alignment concrete, DeterminationProcedure inherits from Procedure to make OMS 3.0 alignment concrete.

A.4. Why Explicit Domain Modeling

The alternative. One could model everything with generic Features and Observations — Sources as plain Features, emission data as Observation results, and relationships as ad-hoc properties. This approach uses only existing OGC standards without defining new classes.

The problem. Generic modeling loses domain semantics. Without EmissionML's explicit classes, systems cannot:

- distinguish a Source from any other Feature,
- enforce that an emission has a Mechanism and an Intent,
- trace the evidence chain from Observation to Emission Quantity to Emission Event,
- validate that required relationships are present (e.g., every Emission Event must have a Source, a Mechanism, and a Determination Procedure).

What explicit modeling provides:

Interoperability	Systems exchanging EmissionML data share a common vocabulary. EmissionEvent, Source, Mechanism, and EmissionQuantity are unambiguous — they mean the same thing in every implementation.
Validation	Requirements can be formally specified and enforced. For example, Requirement 1 mandates that every Emission Event has a Source, a Mechanism, start time, quantity, and their corresponding Determination Procedures.
Usability	Domain experts work with emission concepts directly — *emission events*, *sources*, *mechanisms* — rather than mapping to and from generic abstractions. This reduces the cognitive burden and the risk of misinterpretation.

Analogy. Just as CityGML defines explicit Building, Road, and Vegetation classes rather than using generic Features, EmissionML defines explicit emission domain classes. The General Feature Model provides the abstract foundation; EmissionML provides the domain-specific vocabulary that makes emission data interoperable and actionable.

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ANNEX B (INFORMATIVE) BIBLIOGRAPHY

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